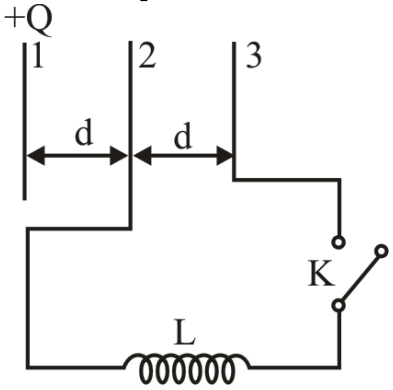
PART-1 : PHYSICS

SECTION-I (i)

1) Three identical fixed metal plates are arranged in the air at equal distances d from each other. The area of each of the plates equals A. The plate 1 has charge + Q, while plates 2 and 3 are not charged and are connected through a key K to the inductor L. The distance between the plates is small compared with their size. The ohmic resistance of the coil can be neglected :-



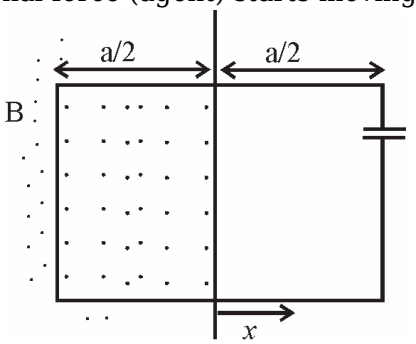
(A) Just before the switch K is closed, the charge on right surface of plate 2 is Q/2

(B) Just before the switch K is closed, the charge on right surface of plate 3 is Q/2

(C) After the switch K is closed, the maximum current in the circuit is $\frac{Q}{2}\sqrt{\frac{d}{L\varepsilon_0 A}}$

(D) After the switch K is closed, the maximum current in the circuit is $Q\sqrt{\frac{d}{L\varepsilon_0 A}}$

2) A square rigid loop of dimension 'a' meter is placed in a region of uniform magnetic field B, as shown. Half of the loop is inside & other half is outside the field. Self inductance of the loop is L = 100 henry & the resistance of loop is $R = 50\Omega$. A capacitor is also connected in loop in one of the side of loop having capacitance C = 1 μF. At t = 0 sec, external force (agent) starts moving loop according to equation $x = \frac{a}{2}\sin(\omega_0 t)$, where $\omega_0 = 100$ rad/s.



(A) Current starts with anticlockwise direction, after some time its direction may change

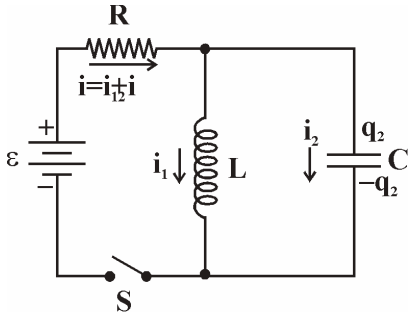
(B) Peak value of current in loop is $\left(\frac{B\omega_0 a^2}{2R}\right)$

Average power delivered by external agent in one cycle is

(C) $\left(\frac{B^2\omega_0^2a^2}{8R}\right)$

(D) If angular frequency of oscillation is increased from ω_0 then average power delivered by external agent will decrease

3) Consider the circuit shown in figure. At time t = 0, switch S is closed. The current through the inductor is i_1, the current through the capacitor branch is i_2, and the charge on the capacitor is q_2.



Then choose the **CORRECT** statements :-

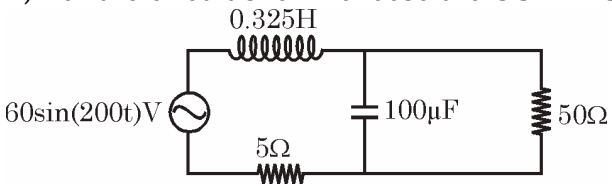
(A) At any instant according to Kirchoff rule, $R(i_1 + i_2) + L\left(\frac{di_1}{dt}\right) = \varepsilon$

(B) At any instant according to Kirchoff rule, $R(i_1 + i_2) - \frac{q_2}{C} = \varepsilon$

(C) Initial values of i_1 & i_2 will be zero and $\frac{\varepsilon}{R}$ respectively

(D) Initial values of i_1 & i_2 will be $\frac{\varepsilon}{R}$ and zero respectively

4) For the circuit shown choose the **CORRECT** option(s) :-



(A) the potential difference across the inductor is 78 cos (200t – 37°)V

(B) the potential difference across the inductor is 78 sin (200t + 37°)V

(C) the effective current through the AC source is 1.2 A

(D) the effective current through the AC source is 0.6 $\sqrt{2}$ A

5) Angular displacement of an object moving on a circle of radius 2m is given by $\theta = \frac{\pi}{6}t + \frac{\pi}{6}t^2$ where θ is in radian and t is in s. For a time duration of t = 0 to t = 1s :

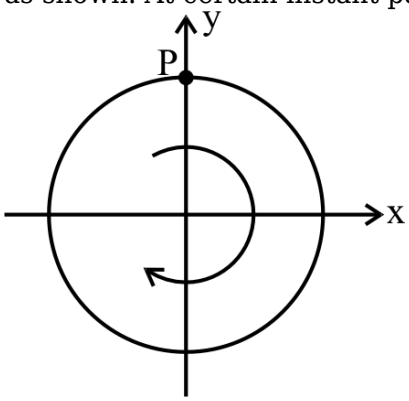
(A) Average angular velocity is $\frac{\pi}{3}$ rad/s

(B) Average angular velocity is $\frac{\pi}{6}$ rad/s

(C) Average velocity is $\frac{3\pi}{3}$ m/s

(D) Average velocity is 2 m/s

6) A particle is moving on a circle of radius 2m in x-y plane as shown. At certain instant particle is at



point P and its acceleration is $6\hat{i}-8\hat{j}$. Then at this moment.

(A) Angular acceleration = $-3\hat{k}$

(B) Velocity = $-4\hat{i}$

(C) Angular velocity = $-2\hat{k}$

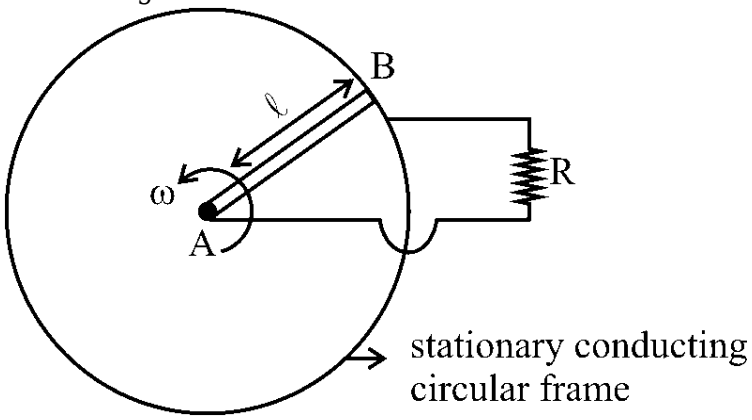
(D) Centripetal acceleration = $-8\hat{j}$

SECTION-I (ii)

Common Content for Question No. 1 to 2

If a conductor is accelerated, initially the free electrons lag behind, thus setting up an electric field in the conductor. In steady state this field exerts force on free electrons and make them accelerate along with the conductor. (Historically this was used by Tolman-Stewant in 1916 for their experiments to measure charge to mass ratio).

1) Figure shows a rod AB that can rotate with uniform angular velocity ω over a stationary conducting circular frame and resistance R is connected between centre A and frame.



If the current was found to be I then the charge to mass ratio of electrons would be

(A) $\frac{\omega^2\ell^2}{IR}$

If the current was found to be I then the charge to mass ratio of electrons would be

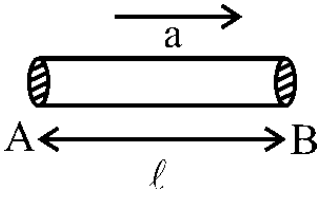
(B) $\frac{2\omega^2\ell^2}{IR}$

If a magnetic field B is applied perpendicular to plane of the paper then induced current (due to

(C) motional emf only) in rod will be $\frac{1}{2}\frac{B\ell^2\omega}{R}$

(D) If a magnetic field B is applied perpendicular and inside the plane of the paper then induced current in rod will be radially outward

2) If the above rod accelerates with acceleration $\vec{a}$ along rod length and v_A and v_B are potential of



end A and B respectively, then

(A) Potential of point A is equal to that of B

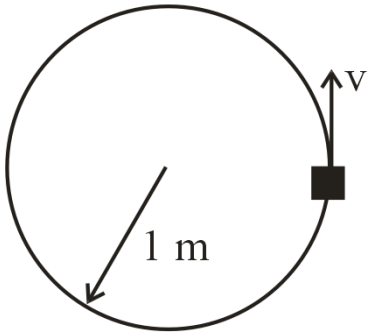
(B) Potential of point B is less than that of A

(C) A constant electric field appears in the rod from B to A

(D) A constant electric field appears in the rod from A to B

Common Content for Question No. 3 to 4

A toy car is moving on a horizontal circular track of radius R. Speed of car is v. Mass of toy is 2 kg. Match the Column-I with Column-II, as per the specifics provided in Questions 3 and 4.



Column-I		**Column-II**	
(P)	Rate of change of speed at t = 1 sec in m/s^2	(1)	1
(Q)	Magnitude of rate of change of velocity at t = 1 sec in m/s^2	(2)	$\sqrt{2}$
(R)	Friction force acting on car at t = 1 sec in newton	(3)	2
(S)	Angle rotated by car in first two sec in radian	(4)	$2\sqrt{2}$
		(5)	$4\sqrt{2}$
		(6)	16/3

3) $v = t^2$ m/s where t is time in seconds and R =0.5 m

(A) P→2, Q→3, R→4, S→5

(B) P→3, Q→4, R→5, S→6

(C) P→1, Q→2, R→3, S→6

(D) P→3, Q→4, R→6, S→4

4) v= t m/s, where t is time in seconds and R = 1m.

(A) P→2, Q→2, R→5, S→3

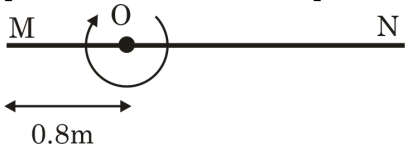
(B) P→2, Q→2, R→4, S→6

(C) P→1, Q→2, R→4, S→3

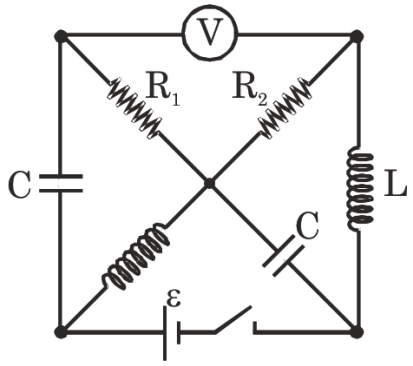
(D) P→1, Q→2, R→3, S→ 4

SECTION-II

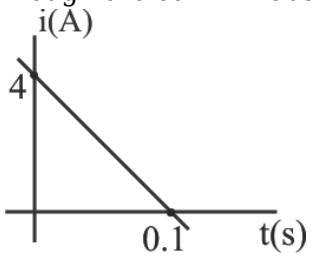
1) A straight conducting wire MN of length 4m is rotated in a uniform magnetic field $\vec{B} = 0.1\hat{k}$ Tesla about point O with angular speed ω = 10 rad/s as shown in the figure. The axis of rotation is parallel to $\vec{B}$. Find the potential difference (in volt) between point M and point N.



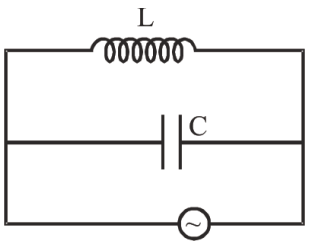
2) In the circuit shown below we first put the switch on and wait for the currents to come to a steady state. Then we put the switch back off. What is the magnitude of the potential difference (in volts) measured by the voltmeter immediately after the switch was turned off ? Take the capacitors and coils to be ideal and assume that the voltmeter provides an infinite resistance. Put ε = 1.25V and R_1 = $4R_2$.



3) Some magnetic flux is changed from a coil of resistance 10Ω. As a result an induced current is developed in it which varies with time as shown in figure. Find the magnitude of change in flux through the coil in weber.

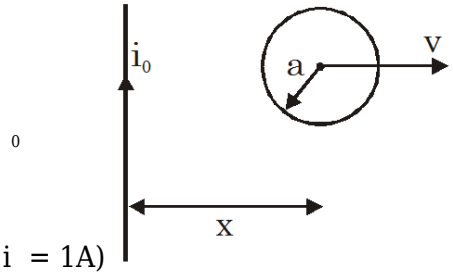


4) An alternating voltage V_0 = 100 V with angular frequency ω is connected across the capacitor and inductor having X_L = 2Ω and X_C = 10Ω. Find the ratio of current through inductor to AC source.



5) At t_1 = 2.00s, the acceleration of a particle in counter clockwise circular motion is $\left(4.00\,m/s^2\right)\hat{i} + \left(3.00m/s^2\right)\hat{j}$. It moves at constant speed. At time t_2 = 5.00s, the particle's acceleration is $\left(3.00\,m/s^2\right)\hat{i} + \left(-4.00m/s^2\right)\hat{j}$.
What is the radius (in m) of the path taken by the particle if (t_2 – t_1) is less than one period? (π^2 = 10).

6) A small conducting loop of radius a and resistance r is pulled with velocity v perpendicular to a long straight conductor carrying a current i_0. If a constant power P is dissipated in the loop, find the velocity (in m/s) of the loop when x = 1m. Given that x >> a. (Given : P = 1 × 10^{-15}W, R = 1mΩ, a =



1cm, i = 1A)

7)

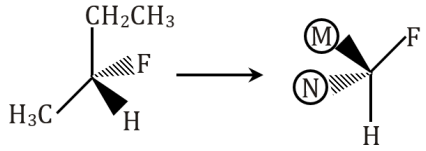
A cyclist speeding at 2 m/s on a level road takes a sharp circular turn of radius 2m without reducing the speed. If minimum coefficient of static friction between tyres and road so that cyclist does not slip while taking the turn is $\frac{1}{n}$, find value of n. (Take g = 10 m/s^2)

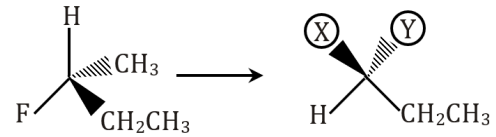
8) In a uniform gravitational field g, which is acting vertically downward, a ball is thrown at an angle with horizontal such that the initial (immediately after projection) radius of curvature is 8 times the minimum radius of curvature. If angle of projection is θ°, then find the value of θ (in degree). (g = 10 m/s^2)

PART-2 : CHEMISTRY

SECTION-I (i)

1) Re-orient the molecule at the left to match the partially drawn perspective at the right. Find the two missing substituents at their correct positions.

(A) 

(B) 

(A) $M=CH_3CH_2-$
$N=CH_3-$

(B) $X=CH_3-$
$Y=F-$

(C) $M=CH_3-$
$N=CH_3CH_2-$

(D) $X=F-$
$Y=CH_3-$

2) The correct combination of names for isomeric alcohols with molecular formula $C_4H_{10}O$ is/are-

(A) tert-butanol and 2-methylpropan-2-ol
(B) tert-butanol and 1, 1-dimethylethan-1-ol
(C) n-butanol and butan-1-ol
(D) isobutyl alcohol and 2-methylpropan-1-ol

3) Select the incorrect statement(s) :

(A) In K_2MnO_4 and CrO_2Cl_2, The central transition metals/ions have the same oxidation state.
(B) $K_2Cr_2O_7$ does not act as oxidiser in acidic medium.
(C) Potassium dichromate on strong heating evolves oxygen gas and forms green powder.
(D) Potassium permangnate on heating with solid KOH evolves oxygen gas and forms a black powder.

4) Select the **CORRECT** statements :

(A) The smaller the ligand, the closer it comes to the metal ion and thus the greater the repulsion between ligand and the electrons present in the d-orbital of central atom that is why $I^- < Br^- < Cl^- < F^-$ are present in spectrochemical series as increasing ligand strength order.

(B) Aminoethanethiolate complex of Pt(II), $[Pt(NH_2CH_2CH_2S)_2]$ has no enantiomers but has geometrical isomers while aminoethanethiolate complex of cobalt(II), $[Zn(NH_2CH_2CH_2S)_2]$ has no enantiomers and no geometrical isomers.

(C) $[Ni(H_2O)_6]^{2+}$ is green, $[Ni(NH_3)_6]^{2+}$ is deep blue and $[Ni(CN)_4]^{2-}$ is yellow because $[Ni(H_2O)_6]^{2+}$ is absorbed red colour, $[Ni(NH_3)_6]^{2+}$ is absorbed orange colour and $[Ni(CN)_4]^{2-}$ is absorbed violet indigo colour.

(D) $Fe(SCN)_3$ with oxalates form optically inactive , colourless complex.

5) Co-ordination entity — Wave length of light absorbed (nm)

$[CoCl(NH_3)_5]^{2+}$	P
$[Co(NH_3)_5(H_2O)]^{3+}$	Q
$[Co(NH_3)_6]^{3+}$	R
$[Co(CN)_6]^{3-}$	S
$[Cu(H_2O)_4]^{2+}$	T

Select the correct orders.

(A) P > R > Q

(B) T > P > Q > R > S

(C) P > Q > R

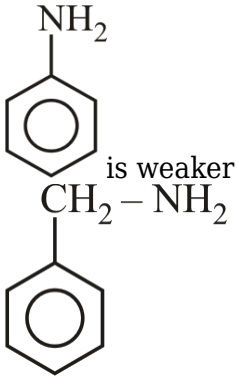
(D) T > Q > P > R > S

6) Which of the following statement(s) is/are true :

(A) $\overset{\ominus}{N}H_2$ is stronger base than $\overset{\ominus}{O}H$

(B) $PhNH_2$ is less basic than NH_3

(C) $CH_3–NH_2$ is weaker base than $N(CH_3)_3$ in (aq) medium

(D)  is weaker base than

SECTION-I (ii)

Common Content for Question No. 1 to 2

The crystal field splitting, Δ_0, depends upon the field produced by the ligand and charge on the metal ion. Some ligands are able to produce strong fields in which case, the splitting will be large whereas others produce weak fields and consequently result in small splitting of d-orbitals.
The influence of the ligand on the colour of a complex may be illustrated by considering the $[Ni(H_2O)_6]^{2+}$ complex (which forms when nickel(II) chloride is dissolved in water) and reaction of it with ethane 1, 2-diamine (en).

1) If the bidentate ligand ethane-1,2-diamine (en) is progressively added in green aqueous solution of $[Ni(H_2O)_6]Cl_2$ in the molar ratios en : Ni as, 1 : 1, 2 : 1, 3 : 1 then their associated colours are

respectively :

(A) pale blue, violet, blue

(B) violet, pale blue, blue

(C) violet, blue, pale blue

(D) pale blue, blue, violet

Common Content for Question No. 2 to 3

MnO_2 is the most important oxide of maganese. It occurs naturally as the black coloured mineral pyrolusite. It is an oxidising agent, and decomposes to Mn_3O_4 on heating to 530°C. It is used in the preparation of potassium permanganate and in the production of Cl_2 gas. Over half a million tonnes per year of MnO_2 is used in dry batteries.

2) In which of the following species, the colour is due to charge transfer.

(I) $[Mn(OH)_4]^{2\ominus}$ (II) $MnO_4^{2\ominus}$

(III) MnO_2 (IV) $KMnO_4$

(A) I, II, III

(B) II, IV

(C) I, II

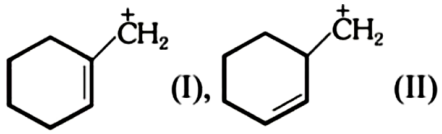
(D) Only IV

Common Content for Question No. 3 to 4

Carbocation is a specie with positively charged carbon atom having six electrons in the valence shell after sharing. Carbocations are formed in the heterolysis of a bond and are planar species. Stability of carbocation is determined by inductive effect, hyperconjugation and resonance effect. Greater tha number of contributing structures, more is the stability of a Carbocation. Electron releasing groups (+ I effect) increases the stability of a carbocation whereas the electron withdrawing groups (– effect) have an opposite effect.

3) In which of the following cases, the carbocation (I) is less stable than the carbocation (II) ?

(A) $C_6H_5-\overset{+}{C}H_2\,(I)\,,\,CH_2=CH-\overset{+}{C}H_2\,(II)$

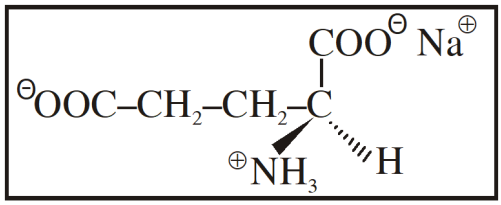
(B) 

(C) $CH_2=\overset{+}{C}H\,(I)\,,CH_3-\overset{+}{C}H_2\,(II)$

(D) $H_3C-\overset{+}{C}H_2\,(I),\ \underset{\underset{F}{|}}{CH_2}-\overset{\oplus}{C}H_2\ (II)$

Common Content for Question No. 4 to 5

S(+) Mono sodium Glutamate (MSG) is a flavour enhancer used in many foods. Fast foods often contain substantial amount of MSG and is widely used in Chinese food. If one mole of above MSG was placed in 845 ml solution and passed through 200 mm tube, the observed rotation was found to

be + 9.6°. 

4) Find out the approximate percentage composition of (–) MSG in a mixture containing (+) MSG and (–) MSG whose specific optical rotation is –20° :

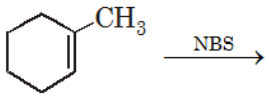
(A) 83.3%

(B) 16.7%

(C) 91.6%

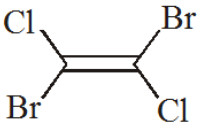
(D) 74%

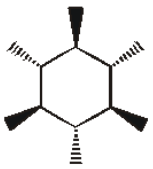
SECTION-II

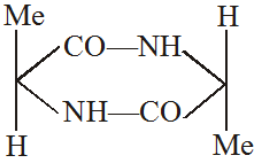
1)  Number of monobromoproducts formed (including stereoisomerism)

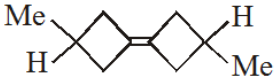
2) How many of the following having POS (Plane of symmetry) and COS (center of symmetry) (both)

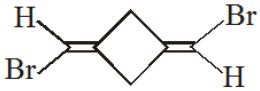
(a) 

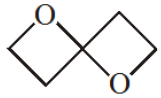
(b) 

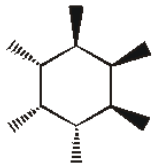
(c) 

(d) 

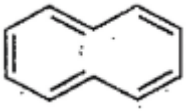
(e) 

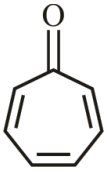
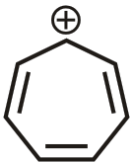
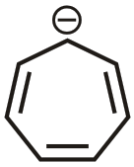
(f) 

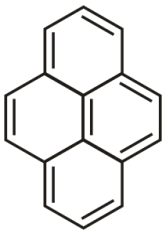
(g) 

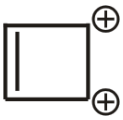
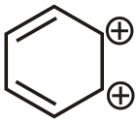
(h) 

3) Find out number of aromatic compounds or ion from following.

(i)  (ii)  (iii)  (iv) 

(v)  (vi)  (vii)  (viii) 

(ix)  (x)  (xi)  (xii) 

(xiii)  (xiv) 

4) For an octahedral complex of Fe^{3+} the value of C.F.S.E. (Crystal field stabilization energy) is "$x\Delta_0$" if magnetic moment of complex is $\sqrt{35}$ B.M. Find out the value of |x|. (Δ_0 is crystal field splitting energy in octahedral field)

5)

Find the total number of ligand(s) which is/are symmetrical bidentate.

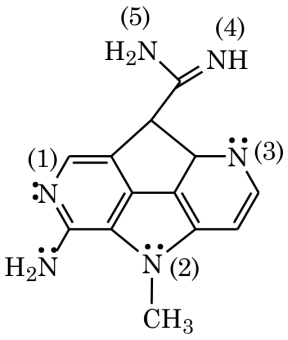
N_2H_4, en, $C_2O_4^{-2}$, gly, Py, dipy,

Phen, Dien, and PPh_3

6) When $KMnO_4$ crystals are added into conc. H_2SO_4 then finally green oily liquid is produced. Find the maximum number of identical bond lengths in it.

7) The complex [Pt(Py)(NH_3)BrCl] will have how many stereo isomers.

8) Sum of tags of Nitrogen lone pairs which are delocalized.



PART-3 : MATHEMATICS

SECTION-I (i)

1) If cos x + cos y = a, cos 2x + cos 2y = b, cos 3x + cos 3y = c, then

(A) $\cos^2 x + \cos^2 y = 1 + \frac{b}{2}$

(B) $\cos x \cos y = \frac{a^2}{2} - \left(\frac{b+2}{4}\right)$

(C) $2a^3 + c = 3a(1 + b)$

(D) $a + b + c = 3abc$

2) If A, B, C are angle of a triangle ABC and tanA.tanC = 3; tanB.tanC = 6; then which of the following is/are **correct** ?

(A) $A = \frac{\pi}{4}$

(B) tanA.tanB = 2

(C) $\frac{\tan A}{\tan C} = 3$

(D) tanB = 2tanA

3) The equation sinx + cos(k + x) + cos(k – x) = 2 has real solution(s), then sink can be

(A) $\frac{-3}{4}$

(B) $\frac{1}{4}$

(C) $\frac{1}{2}$

(D) $\frac{3}{4}$

4) If m > 0, n < 5, 0 < m + n < 10 and $\frac{x^2+5}{2}$ = x – 2 cos(m + nx) has atleast one real root, then

(A) the greatest value of (m + n) is 3π.

(B) the greatest value of (m + n) is 2π.

(C) the least value of (m + n) is π.

(D) the least value of (m + n) is 2π.

5) cos 15x = sin 5x if

(A) $x = -\frac{\pi}{20} + \frac{n\pi}{5}, n \in I$

(B) $x = \frac{\pi}{40} + \frac{n\pi}{10}, n \in I$

(C) $x = \frac{3\pi}{20} + \frac{n\pi}{5}, n \in I$

(D) $x = -\frac{3\pi}{40} + \frac{n\pi}{10}, n \in I$

6) Given that A, B are positive acute angles and $\sqrt{3}\sin 2A = \sin 2B$ & $\sqrt{3}\sin^2 A + \sin^2 B = \frac{\sqrt{3}-1}{2}$, then A or B may take the value(s) -

(A) 15°

(B) 30°

(C) 45°

(D) 75°

SECTION-I (ii)

Common Content for Question No. 1 to 2

An operator Δ is defined to operate on differentiable functions defined as follows.

If ƒ (x) is a differentiable function then $\Delta(f(x)) = \lim_{h\to 0} \frac{f^3(x+h) - f^3(x)}{h}$

g(x) is a differentiable function such that the slope of the tangent to the curve y = g(x) at any point (a, g(a)) is equal to $2e^a$ (a+1) also g(0)=0.

On the basis of above information, answer the following questions

1) $\Delta(\Delta(x + 2))_x = 0$

(A) $2^5 \cdot 3^9$

(B) $2^9 \cdot 3^5$

(C) $2^4 \cdot 3^5$

(D) $2^6 \cdot 3^4$

Common Content for Question No. 2 to 3

An operator Δ is defined to operate on differentiable functions defined as follows.

If ƒ (x) is a differentiable function then $\Delta(f(x)) = \lim_{h\to 0} \frac{f^3(x+h) - f^3(x)}{h}$

g(x) is a differentiable function such that the slope of the tangent to the curve y = g(x) at any point (a, g(a)) is equal to $2e^a$ (a+1) also g(0)=0.

On the basis of above information, answer the following questions

2) $\lim_{x\to 0}\frac{\Delta g(x)}{\ell n(\cos 2x)}$

(A) 12

(B) –12

(C) 24

(D) –24

Common Content for Question No. 3 to 4

For the curve sin x + siny = 1 lying in the first quadrant there exists a constant α for which

$\lim_{x\to 0} x^{\alpha}\frac{d^2y}{dx^2} = L$ (not zero)

3) The value of L :

(A) $\frac{1}{2}$

(B) 1

(C) $\frac{1}{2\sqrt{2}}$

(D) $\frac{1}{2\sqrt{3}}$

Common Content for Question No. 4 to 5

For the curve sin x + siny = 1 lying in the first quadrant there exists a constant α for which

$\lim_{x\to 0} x^{\alpha}\frac{d^2y}{dx^2} = L$ (not zero)

4) The value of α

(A) $\frac{1}{2}$

(B) $\frac{1}{\sqrt{2}}$

(C) $\frac{3}{2}$

(D) 2

SECTION-II

1) Let f be a twice differentiable function with f(–1) = 1 = f(1) and f(0) = 0. then the minimum number of roots of equation $f(x)(f'(x))^3 + (f(x))^2 f'(x) f''(x) - f(x) f'(x) = x(f'(x))^2 + x f(x) f''(x) - x$ is

2) If Ax + By = 1 is a normal to $a^5y = x^6$, then $(mA)^m = Ba^{m-1}(mA^2 + B^2)^{m-1}$ where m is

3) If x is real, then the maximum value of $y = 2(3-x)\left(x + \sqrt{x^2+4}\right)$ is

4) If $P(x)$ be a polynomial of degree 3 satisfying $P(-1) = 10, P(1) = -6$ and $P(x)$ has maximum at $x = -1$ and $P'(x)$ has minima at $x = 1$. Find the distance between the local maximum and local minimum of the curve.

5) Let f : [– 1, 1] → R be defined as $f(x) = ax^2 + bx + c$ for all x ∈ [–1, 1], where a, b, c ∈ R such that f(–1) = 2, f'(–1) = 1 and for x ∈ (–1, 1) the maximum value of f''(x) is $\frac{1}{2}$. If f(x) ≤ a, x ∈ [–1, 1], then the least value of a is equal to

6) If y = mx + c is tangent (a line is tangent to the curve at two distinct points) on the curve $y = x^4 + 2x^3 - 11x^2 - 13x + 35$, then the value of $\frac{|m+c|}{5}$ is
If tangent to the curve $xy = x^2 + 1$ at (α, β) is normal to the curve $x^2 + y^2 + 2fy + c = 0$, then |f.α| is

7) The point of contact of the tangent to the curve $x^{2017}y^{1033} = K$ divides the portion of the tangent between the co-ordinate axes in the ratio of a : b (a from x-axis), then sum of the digits of (a + b) is

8) Let $4\left(\frac{ay}{\cos\alpha} - \frac{bx}{\sin\alpha}\right) = 3(a+b),$ ab ≠ 0 is normal to the curve $x = 3\cos\theta - \cos^3\theta$, $y = 3\sin\theta - \sin^3\theta$ at θ = α, then number of possible values of α, $\alpha \in (0,\pi) - \left\{\frac{\pi}{2}\right\}$ is

ANSWER KEYS

PART-1 : PHYSICS

SECTION-I (i)

Q.	1	2	3	4	5	6
A.	A,B,C	A,B,C,D	A,C	B,D	A,D	A,C,D

SECTION-I (ii)

Q.	7	8	9	10
A.	C	C	B	C

SECTION-II

Q.	11	12	13	14	15	16	17	18
A.	4.80	3.75	2.00	1.25	2.00	15.80 to 16.00	5.00	60.00

PART-2 : CHEMISTRY

SECTION-I (i)

Q.	19	20	21	22	23	24
A.	A,B	A,C,D	B,D	A,C	B,C	A,B,D

SECTION-I (ii)

Q.	25	26	27	28
A.	D	B	C	C

SECTION-II

Q.	29	30	31	32	33	34	35	36
A.	9.00	6.00	10.00	0.00	4.00	6.00	3.00	10.00

PART-3 : MATHEMATICS

SECTION-I (i)

Q.	37	38	39	40	41	42
A.	A,B,C	A,B,D	B,C	A,C	A,B,C,D	A,B

SECTION-I (ii)

Q.	43	44	45	46
A.	D	B	C	C

SECTION-II

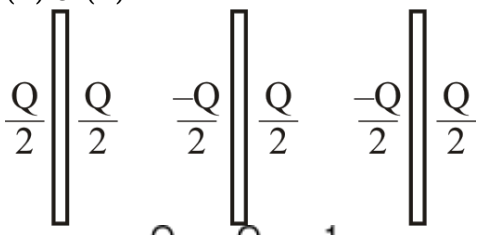
Q.	47	48	49	50	51	52	53	54
A.	**3.00**	**6.00**	**13.00**	**3.00**	**5.00**	**2.00**	**8.00**	**4.00**

SOLUTIONS

PART-1 : PHYSICS

1)

(A) & (B)



(C) $i_{max} = \frac{Q}{2}\omega = \frac{Q}{2}\frac{1}{\sqrt{L\varepsilon_0 A/d}}$

2) $\phi = a\left[\frac{a}{2} - \frac{a}{2}\sin(\omega t)\right]B$

$\frac{d\phi}{dt} = -\omega\frac{Ba^2}{2}\cos(\omega t)$

$\varepsilon = -\frac{d\phi}{dt} = \frac{B\omega a^2}{2}\cos(\omega t)$ (Initially E is anticlockwise positive)

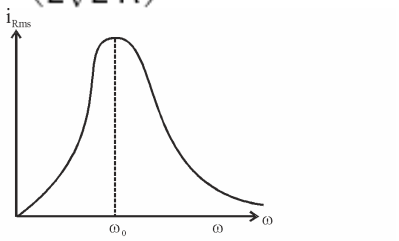
$i_{rms} = \frac{1}{\sqrt{2}}\frac{B\omega a^2/2}{\sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}} = \frac{B\omega a^2}{2R}\cdot\frac{1}{\sqrt{2}}$

as $= \omega = \frac{1}{\sqrt{LC}}$

$\omega = \omega_0$ is condition of resonance

$P_{avg} = i_{rms} V_{rms} \cos\phi$

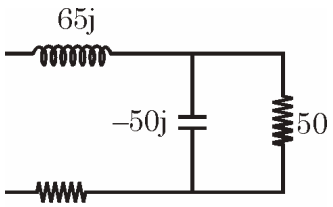
$= \left(\frac{B\omega a}{2\sqrt{2}R}\right)^2 R = \frac{B^2\omega^2 a^2}{8R}$



Graph implies that when $\omega > \omega_0$, I_{Rms} decreases in circuit so power decreases.

3) At t = 0, inductor behaves as open circuit and capacitor as short circuit.

and in (B), by KVL, $R(i_1 + i_2) + \frac{q_2}{C} = \varepsilon$



4)

$$\tilde{z} = \frac{50(-50j)}{50-50j} + 5 + 64j = 30 + 40j$$

$$\tilde{i} = \frac{60e^{j\omega t}}{30+40j}$$

$$\tilde{v}_L = \tilde{i}\tilde{x}_L = \frac{60}{30+40j}(65j)\,e^{j\omega t}$$

$$= 78\left(\frac{4}{5} + \frac{3}{5}j\right)e^{j\omega t}$$

$$= 78e^{j37^\circ}e^{j\omega t}$$

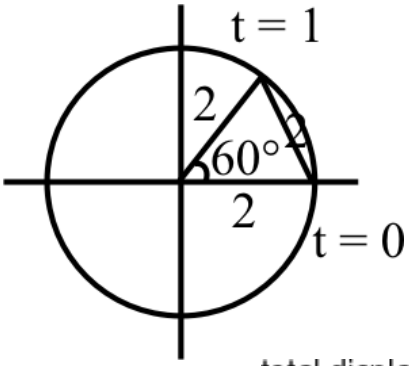
$$= 78ej(wt + 37°)$$

5)

$$\text{Average angular velocity} = \frac{\text{total angular displacement}}{\text{total time}}$$

$$= \frac{\left(\frac{\pi}{6} + \frac{\pi}{6}\right) - 0}{1}$$

$$< \omega > = \frac{\pi}{3}$$



$$\text{average velocity} = \frac{\text{total displacement}}{\text{total time}}$$

$$< \vec{v} > = \frac{2}{1} = 2\text{m/sec}^2$$

6)

$a_t = 6$

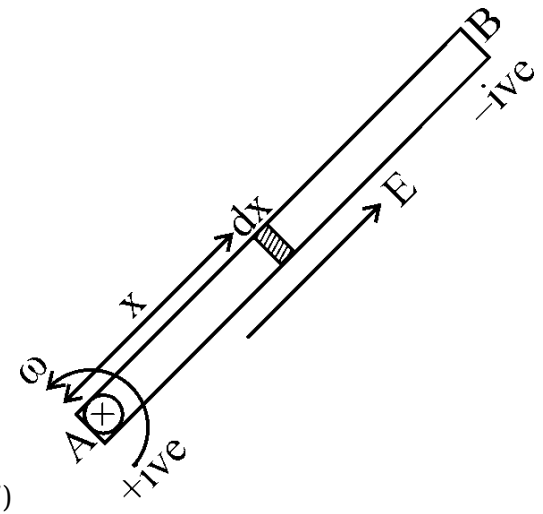
$a_t = R\alpha$

$6 = 2\alpha$

$a_c = 8$

$a_c = R\omega^2$

$\omega = 2$



7)

Due to centrifugal force electrons are moving from A to B.

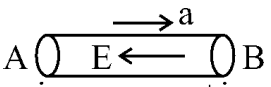
At equilibrium, Electric force due to induced charges become equal to centrifugal force $m\omega^2x$ = eE

$$\int_0^{\ell} m\omega^2 x dx = \int_0^{\ell} eEdx = e\int_0^{\ell} Edx$$

$$\Rightarrow \frac{m\omega^2\ell^2}{2} = ev \quad \text{(v = voltage across rod)}$$

$$\Rightarrow \quad v = iR = \frac{m\omega^2\ell^2}{2e} \Rightarrow \frac{e}{m} = \frac{\omega^2\ell^2}{2iR}$$

8) Due to pshedo force, electrons move from B to A.



$\Rightarrow \quad v_B > v_A$

9)

$v = t^2$ m/s

$\frac{dv}{dt} = (2t)$

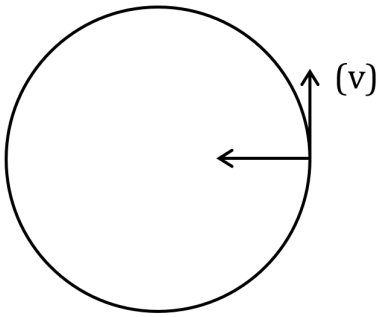
$a_T = 2t = 2$ m/s^2

$a_R = \frac{t^4}{R} = 2$m/s^2

$a_N = \left(2\sqrt{2}\right)$ m/s^2

$$\omega = \frac{t^2}{0.5} = 2t^2 = \frac{d\theta}{dt}$$

$$\int_0^2 2t^2 dt = \frac{2t^3}{3} = \frac{2\times 8}{3} = \left(\frac{16}{3}\right)$$



10)

(v = t m/s)

$a_T = (\alpha R) = (1)$

$$\frac{dv}{dt} = 1 m/s$$

$$\frac{d\vec{v}}{dt} = \sqrt{\left(\frac{v^2}{R}\right)^2 + (1)^2}$$

$$= \sqrt{(1)^2 + 1^2} = \sqrt{2}\ m/s$$

$$f = m(a_T)$$

$$f = 2\sqrt{2}$$

$$\frac{d\theta}{dt} = t$$

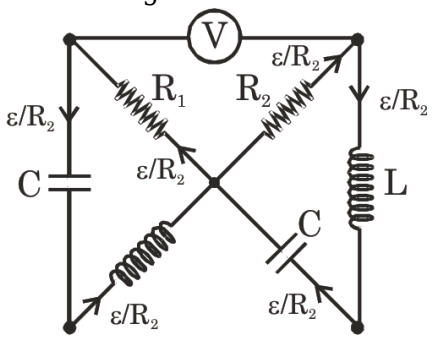
$$d\theta = \int_0^2 t\,dt$$

$$= \frac{t^2}{2} = 2 rad$$

11) $e_1 = \frac{B\omega\ell_1^2}{2},\ e_2 = \frac{B\omega\ell_2^2}{2}$

$\Rightarrow \quad V_N - V_M = e_2 - e_1 = \frac{B\omega}{2}\left(\ell_2^2 - \ell_1^2\right) = 4.80$ volts

12) Initially when the switch is on and the currents reach steady state, the capacitors and voltmeter behaves as elements with infinite resistance. Where as the coil behaves like a conductor with no resistance. Hence the currents through R_1 and R_2 are $I_1 = 0$ and $I_2 = \varepsilon/R_2$ respectively. Immediately after the switch is turned off, due to the property of capacitor and inductor, the potential difference across capacitor and current through coil doesn't change. So circuit diagram becomes



So potential difference across voltmeter is

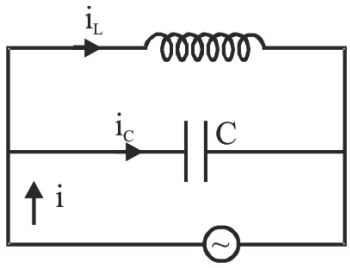
$$= \left|\frac{\varepsilon}{R_2}R_1 - \frac{\varepsilon}{R_2}R_2\right| = 3\varepsilon$$

= 3.75V.

13) $Q = \int i\,dt$ = Area = $\frac{1}{2}(4)(0.1)$ = 0.2 C

$Q = \frac{\Delta\phi}{R}$

$\Delta\phi$ = QR = (0.2) (10) = 2 wb

14) 

If V = V_0 sin ωt

$i_c = \frac{V_0}{X_c}\sin\left(\omega t + \frac{\pi}{2}\right)$

$i_L = \frac{V_0}{x_L}\sin\left(\omega t - \frac{\pi}{2}\right)$

$i = i_C + i_L$

So required ratio = 1.25

15) $5 = \frac{v^2}{R}$

$\left(\sqrt{5R}\right)3 = \frac{3\pi}{2} \times R$

$5R = \frac{10}{4} \times R^2$

$R = \frac{20}{10} = 2m$

16) $\phi = \vec{B} \cdot \vec{A}$

$B = \frac{\mu_0 i}{2\pi x}$, A = πa^2

$\phi = \frac{\mu_0 i a^2}{2x}$

$|\varepsilon| = \frac{d\phi}{dt} = \frac{\mu_0 i a^2}{2}\frac{1}{x^2}\left(\frac{dx}{dt}\right)$

$\varepsilon = \frac{\mu_0 i a^2}{2x^2} \cdot v$

$P = \frac{\varepsilon^2}{R} = \frac{\mu_0^2 i^2 a^4 v^2}{4x^4\,(R)}$

Put P = 10^{-15}, i = 1A, a = 0.01 m, x = 1m,

R = 10^{-3} Ω, $\mu_0 = 4\pi \times 10^{-7}$

We get v = 15.91 m/s

17) $\mu mg = \frac{mv^2}{R}$

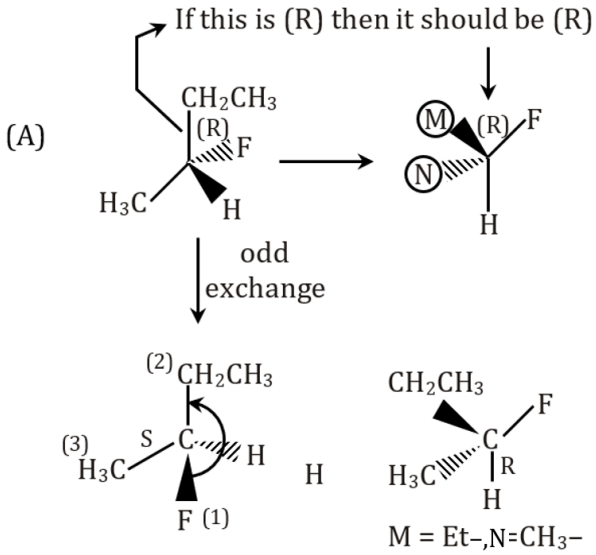
18) Minimum radius of curvature will be at highest point.
Therefore

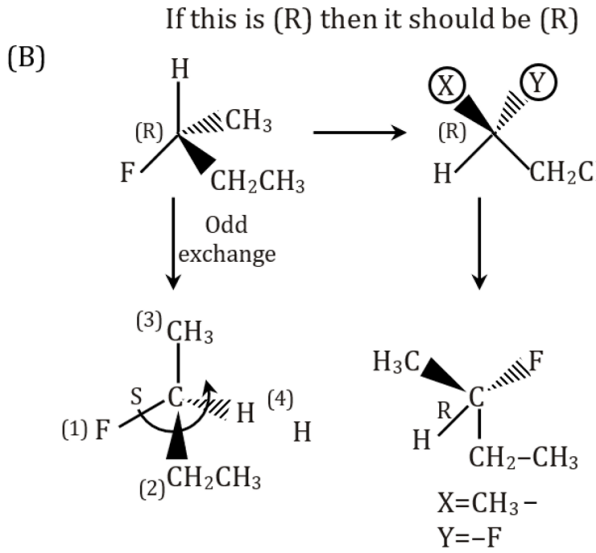
$$\frac{u^2}{g\cos\theta} = 8\frac{u^2\cos^2\theta}{g}$$

$$\Rightarrow \cos\theta = \frac{1}{2} \Rightarrow \theta = 60°$$

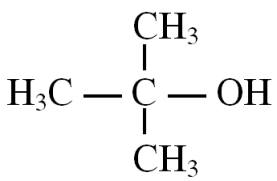
PART-2 : CHEMISTRY

19)



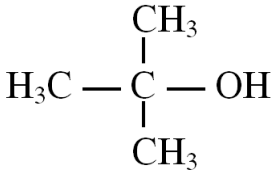


20) **Option 1 :**



correct

tert-Butanol and 2-methylpropan-2-ol

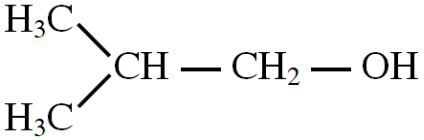


Option 2 : incorrect

tert-Butanol and 1, 1-Dimethylethan-1-ol

Option 3 : $CH_3 — CH_2 — CH_2 — CH_2 — OH$ correct

n-Butanol and Butan-1-ol



Option 4 : correct

Isobutyl alcohol and 2-methylpropan-1-ol

21) **Question Explanation:**

We have to choose incorrect statement.

Concept:

This question is based on chemical properties of $KMnO_4$ and $K_2Cr_2O_7$.

Solution:

(2) $\underset{Cr^{+6}}{Cr_2O_7^{2-}} + 14H^+ + 6e^- \rightarrow \underset{Cr^{+3}}{2Cr^{3+}} + 7H_2O$

Cr it self converts to Cr^{+3} from Cr^{+6}, i.e. reduces and oxidises others like I^-, Sn^{2+}, H_2S, Fe^{2+}

(4) $2KMnO_4 \rightarrow K_2MnO_4 + \underset{\text{Black Brown}}{\underset{\downarrow}{MnO_2}} + O_2$

Final Answer:

The correct option is (2) & (4).

22) $[Zn(NH_2CH_2CH_2S)_2]$ has enantiomer pair and no geometrical isomers due to tetrahedral shape $[Fe(C_2O_4)_3]^{3-}$ ion has sp^3d^2 hybridisation outer orbital complex and high spin, optically active and coloured compound

23)

Coordination entity	**Wavelength of light absorbed (nm)**	
$[CoCl(NH_3)_5]^{2+}$	(P)	535
$[Co(NH_3)_5(H_2O)]^{3+}$	(Q)	500
$[Co(NH_3)_6]^{3+}$	(R)	475
$[Co(CN)_6]^{3-}$	(S)	310
$[Cu(H_2O)_4]^{2+}$	(T)	600

24)

Order of basic strength in aq. medium is
$(CH_3)_2\ NH > CH_3NH_2 > (CH_3)_3N > NH_3$

25)

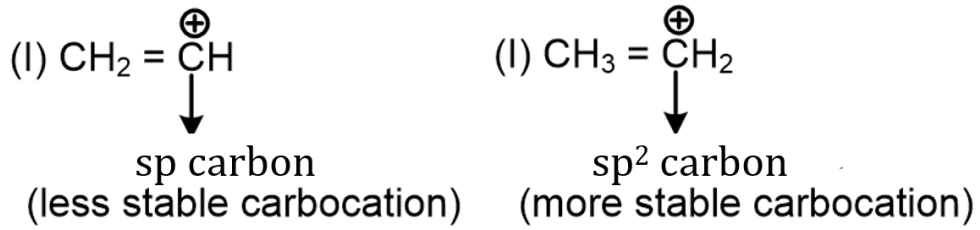
$$[Ni(H_2O)_6]^{2+}(aq.) + en(aq.) \longrightarrow \underset{\text{Pale blue}}{[Ni(H_2O)_4(en)]^{2+}(aq.)} + 2H_2O$$

$$[Ni(H_2O)_4(en)]^{2+}(aq.) + en(aq.) \longrightarrow \underset{\text{Blue/purple}}{[Ni(H_2O)_2(en)_2]^{2+}(aq.)} + 2H_2O$$

$$[Ni(H_2O)_2\ (en)_2]^{2+}(aq.) + en(aq.) \longrightarrow \underset{\text{Violet}}{[Ni(en)_3]^{2+}(aq.)} + 2H_2O$$

26) Cause of colour in $MnO_4^{2\ominus}$ and $KMnO_4$ is L → M charge transfer.

27)



28)

$$[\alpha]_{pure} = \frac{[\alpha]_{obs}}{l.c.} = \frac{+9.6}{2 \times \left(\frac{169}{845}\right)}$$

$[\alpha]_{pure} = +24^\circ$

$[\alpha]_{pure}$ [(–) MSG] = –24°

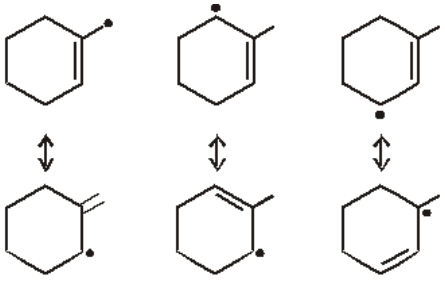
$$ee = \frac{[\alpha]_{mixture}}{[\alpha]_{pure}} \times 100 = \frac{-20}{-24} \times 100 = 83.3$$

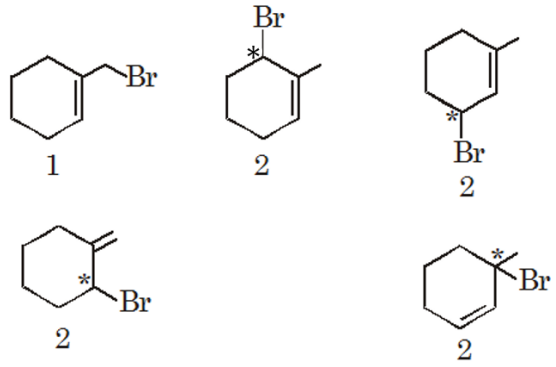
16.7 % → d = 8.35, l = 8.35

 RM = 100 – 83.3 ⇒

Total (–) MSG = 83.3 + 8.35 = 91.6 %

29)



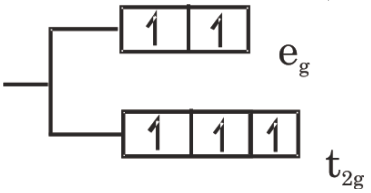


30) Compound which have POS + COS both are → a, b, c, e, f , h

31)

i, iv, v, vi, vii, ix, x, xi, xii, xiii

32) $\mu = \sqrt{35}$
⇒ n = 5 (no. of unpaired e^-)
Fe^{3+} in octahedral field (for n = 5)



$C.F.S.E. = -0.4\Delta_0 n_{t_{2g}} + 0.6\Delta_0 n_{e_g} + mp$
$= -0.4\Delta_0 \times 3 + 0.6\Delta_0 \times 2 + 0$
$= 0$

33)

N_2H_4 → Monodentate

en → Symmetrical bidentate

$C_2O_4^{-2}$ → Symmetrical bidentate

gly → unsymmetrical bidentate

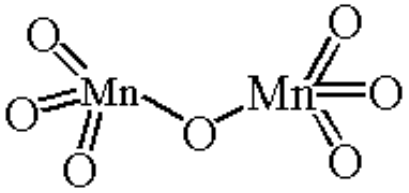
Py→ Monodentate

dipy → Symmetrical bidentate

Phen→ Symmetrical bidentate

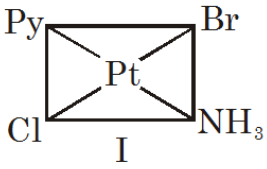
Dien→ Symmetrical tridentate

PPh_3 → Monodentate

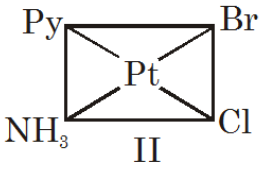


34)

All $Mn = O$

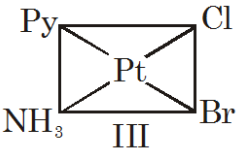


I



II

35)



III

36) Localized lone pairs = 4 and 1
Delocalized lone pairs = 2, 3 & 5

PART-3 : MATHEMATICS

37) $(\cos x + \cos y)^2 = a^2$

$\Rightarrow \cos^2 x + \cos^2 y + 2\cos x \cos y = a^2 \quad ...(1)$

$\cos 2x + \cos 2y = b$

$2\cos^2 x - 1 + 2\cos^2 y - 1 = b$

$2[\cos^2 x + \cos^2 y] = b + 2$

$\Rightarrow \cos^2 x + \cos^2 y = \frac{b}{2} + 1 \quad ...(2)$

From (1) and (2),

$2\cos x \cos y = a^2 - \left(\frac{b+2}{2}\right) \Rightarrow \cos x \cos y = \frac{a^2}{2} - \left(\frac{b+2}{4}\right)$

$\cos 3x + \cos 3y = c$

$4\cos^3 x - 3\cos x + 4\cos^3 y - 3\cos y = c$

$4[\cos^3 x + \cos^3 y] - 3[\cos x + \cos y] = c$

$\Rightarrow 4[(\cos x + \cos y)(\cos^2 x + \cos^2 y - \cos x \cos y)] - 3(\cos x + \cos y) = c$

$\Rightarrow 4\left[a\left(\frac{b+2}{2} - \frac{1}{2}\left(a^2 - \frac{b+2}{2}\right)\right)\right] - 3a = c$

$\Rightarrow 2ab + 4a - 2a^3 + ab + 2a = 3a + c$

$\Rightarrow 2a^3 + c = 3a(1 + b)$

38) tanA tanC = 3 and tanB tanC = 6

Let $\tan A = \alpha$

$\tan B = 2\alpha$

$\frac{\tan A}{\tan B} = \frac{1}{2}$ $\tan C = \frac{3}{\alpha}$

∴ tanA + tanB + tanC = tanA tanB tanC

$\alpha + 2\alpha + \frac{3}{\alpha} = 6\alpha$

$\Rightarrow \quad 3\alpha + \frac{3}{\alpha} = 6\alpha \Rightarrow \frac{3}{\alpha} = 3\alpha$

⇒ α = 1

tanA = 1, tanB = 2, tanC = 3

39) sinx + cos(k + x) + cos(k – x) = 2

⇒ sinx + 2cosk.cosx = 2

we have solution when,

$2 \leqslant \sqrt{1^2 + (2\cos k)^2}$

⇒ 1 + 4cos^{2}k ≥ 4

⇒ $\cos^2 k \geq \frac{3}{4}$

⇒ $1 - \sin^2 k \geq \frac{3}{4}$

⇒ $1 - \frac{3}{4} \geqslant \sin^2 k \Rightarrow \frac{1}{4} \geqslant \sin^2 k$

⇒ $\frac{1}{4} \geqslant \sin^2 k \geqslant 0 \Rightarrow \frac{-1}{2} \leqslant \sin k \leqslant \frac{1}{2}$

For asinx + bcosx = c, have solutions when

As, $|a\sin x + b\cos x| \leqslant \sqrt{a^2 + b^2}$

⇒ $|c| \leqslant \sqrt{a^2 + b^2}$.

40) m > 0; n < 5; 0 < m + n < 10

$\frac{x^2 + 5}{2}$ = x – 2cos(m + nx)

⇒ x^2 – 2x + 5 = –4 cos(m + nx)

⇒ (x – 1)^2 + 4 = –4 cos(m + nx)

Range of (x – 1)^2 + 4 = [4, ∞) and range of

–4 cos(m + nx) = [–4, 4]

Equality 4 = 4 when

(x – 1)^2 = 0 & cos(m + nx) = –1

⇒ x = 1 & cos(m + n(1)) = –1

⇒ m + n = 2nπ + π

= (2n + 1)π

= {π, 3π}

41)

cos 15x = sin 5x

⇒ cos 15x – sin 5x = 0

⇒ cos 10x·cos 5x – sin 10x sin 5x – sin 5x = 0

$\Rightarrow (\cos^2 5x - \sin^2 5x)\cos 5x - \sin 5x\,(1 + \sin 10x) = 0$
$\Rightarrow (\cos^2 5x - \sin^2 5x)\cos 5x - \sin 5x\,(\cos 5x + \sin 5x)^2 = 0$
$\Rightarrow (\cos 5x + \sin 5x)\{(\cos 5x - \sin 5x)\cos 5x - \sin 5x\,(\cos 5x + \sin 5x)\} = 0$
$\Rightarrow (\cos 5x + \sin 5x)\{\cos^2 5x - \sin 5x \cos 5x - \sin 5x \cos 5x - \sin^2 5x\} = 0$
$\Rightarrow (\cos 5x + \sin 5x)(\cos 10x - \sin 10x) = 0$

$\therefore \quad \cos 5x + \sin 5x = 0$

$\Rightarrow \tan 5x = -1$

$5x = n\pi + \frac{3\pi}{4} \quad \therefore \quad x = \frac{n\pi}{5} + \frac{3\pi}{20}, \; n \in I$

or

$5x = n\pi - \frac{\pi}{4} \; \therefore \; x = \frac{n\pi}{5} - \frac{\pi}{20}, \; n \in I$

$\therefore \quad \cos 10x - \sin 10x = 0$

$\Rightarrow \quad \tan 10x = 1$

$10x = n\pi + \frac{\pi}{4} \quad \therefore \quad x = \frac{n\pi}{10} + \frac{\pi}{40}, \; n \in I$

or

$10x = n\pi - \frac{3\pi}{4} \; \therefore \quad x = \frac{n\pi}{10} - \frac{3\pi}{40}, \; n \in I$

42) $\sqrt{3}\sin 2A = \sin 2B$ ..... (1)

$\sqrt{3}\sin^2 A + \sin^2 B = \frac{\left(\sqrt{3}-1\right)}{2}$

$\sqrt{3}2\sin^2 A + 2\sin^2 B = \left(\sqrt{3}-1\right)$

$\sqrt{3}(1-\cos 2A) + (1-\cos 2B) = \sqrt{3}-1$

$\sqrt{3} - \sqrt{3}\cos 2A + 1 - \cos 2B = \sqrt{3} - 1$

$2 - \sqrt{3}\cos 2A = \cos 2B$ ..... (2)

Using (1)

$\sqrt{3}\sin 2A = \sin 2B$

$3\sin^2 2A = \sin^2 2B$

$3\left(1-\cos^2 2A\right) = \left(1-\cos^2 2B\right)$

$2 - 3\cos^2 2A = -\cos^2 2B$

$3\cos^2 2A - 2 - \cos^2 2B = 0$

$3\cos^2 2A - 2 - \left(2-\sqrt{3}\cos 2A\right)^2 = 0$

$3\cos^2 2A - 2 - 4 + 4\sqrt{3}\cos 2A - 3\cos 2A = 0$

$\cos 2A = \frac{6}{4\sqrt{3}}, \cos 2A = \frac{\sqrt{3}}{2}$

$2A = 30°$

$\Rightarrow A = 15º; B = 30º$

43) $\Delta(x+2) = \lim\limits_{h\to 0} \frac{(x+h+2)^3 - (x+2)^3}{h}$

$= \lim\limits_{h\to 0} \frac{(h)\left((x+h+2)^2 + (x+2)^2 + (x+h+2)(x+2)\right)}{h}$

$= 3(x+2)^2$

$\Delta(\Delta(x+2))_{x=0} = \lim\limits_{h\to 0} \frac{27(x+2+h)^6 - 27(x+2)^6}{h}$

$= \lim\limits_{h\to 0} 27\frac{\left\{(2+h)^6 - 2^6\right\}}{h}$

$$= \lim_{h\to 0} 27\frac{\left\{{}^6C_1 2^5 h + {}^6C_2 2^4 h^2 + ... + {}^6C_6 h^6\right\}}{h}$$

$= {}^6C_1 \cdot 3^3 \cdot 2^5$

$= 2^6 \cdot 3^4$

44) $\lim_{x\to 0} \frac{\Delta g(x)(\cos 2x-1)}{\ln(1+(\cos 2x-1))}\frac{4x^2}{(\cos 2x-1)}\cdot\frac{1}{4x2}$

$= \frac{1}{2}\lim_{x\to 0}\frac{\Delta g(x)}{x^2}$

$= -\frac{1}{2}\lim_{x\to 0}\frac{12x^2e^{3x}(2x+2)}{x^2}$

$= -\lim_{x\to 0} 6e^{3x}(2x+2)$

45) sinx + siny = 1

$\Rightarrow \cos x + \cos y\frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = \frac{-\cos x}{\cos y}$

$-\sin x - \sin y\left(\frac{dy}{dx}\right)^2 + \cos y\frac{d^2y}{dx^2}$

$$\frac{d^2y}{dx^2} = \frac{\sin x + \sin y\left(-\frac{\cos x}{\cos y}\right)^2}{\cos y}$$

$$= \frac{\sin x + \frac{\sin y\cos^2 x}{\cos^2 y}}{\cos y}$$

$$\frac{d^2y}{dx^2} = \frac{\sin x + \frac{(1-\sin x)\cos^2 x}{2\sin x-\sin^2 x}}{\sqrt{2\sin x-\sin^2 x}}$$

$$= \frac{2\sin x-\sin^3 x+1-\sin x-\sin^2 x+\sin^3 x}{(2\sin x-\sin^2 x)^{3/2}}$$

$$= \frac{\sin x-\sin^2 x+1}{(\sin x)^{3/2}(2-\sin x)^{3/2}}$$

$x^\alpha\left(\frac{d^2y}{dx^2}\right)$ is finite if $\alpha = \frac{3}{2}$; $L = \frac{1}{2\sqrt{2}}$

46) sinx + siny = 1

$\Rightarrow \cos x + \cos y\frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = \frac{-\cos x}{\cos y}$

$-\sin x - \sin y\left(\frac{dy}{dx}\right)^2 + \cos y\frac{d^2y}{dx^2}$

$$\frac{d^2y}{dx^2} = \frac{\sin x + \sin y\left(-\frac{\cos x}{\cos y}\right)^2}{\cos y}$$

$$= \frac{\sin x + \frac{\sin y \cos^2 x}{\cos^2 y}}{\cos y}$$

$$\frac{d^2y}{dx^2} = \frac{\sin x + \frac{(1-\sin x)\cos^2 x}{2\sin x - \sin^2 x}}{\sqrt{2\sin x - \sin^2 x}}$$

$$= \frac{2\sin x - \sin^3 x + 1 - \sin x - \sin^2 x + \sin^3 x}{(2\sin x - \sin^2 x)^{3/2}}$$

$$= \frac{\sin x - \sin^2 x + 1}{(\sin x)^{3/2}(2 - \sin x)^{3/2}}$$

$x^{\alpha}\left(\frac{d^2y}{dx^2}\right)$ is finite if $\alpha = \frac{3}{2}$; $L = \frac{1}{2\sqrt{2}}$

47) Given expression is
$(f(x)\, f'(x) - x)\,((f'(x))^2 + f(x)\, f''(x) - 1) = 0$
consider $g(x) = f^2(x) - x^2$
since $g(x) = 0$ has 3 real roots
$\Rightarrow g'(x) = 0$ must have minimum 2 real roots
and $g''(x) = 0$ must have minimum 1 real root

48) $y' = \frac{6x^5}{a^5} = \frac{B}{A} \Rightarrow x = a\left(\frac{B}{6A}\right)^{1/5}$

$$y = \frac{a^6}{a^5}\left(\frac{B}{6A}\right)^{6/5} = a\left(\frac{B}{6A}\right)^{6/5}$$

$$A\left(\frac{aB^{15}}{6^{1/5}A^{1/5}}\right) + B\left(\frac{aB^{6/5}}{6^{6/5}A^{6/5}}\right) = 1$$

$\Rightarrow \quad 6a\,A^2B^{1/5} + aB^2B^{1/5} = 6^{6/5}A^{6/5} \quad \Rightarrow \quad a(6A^2 + B^2)B^{1/5} = (6A)^{6/5}$
$\Rightarrow \quad Ba^5\,(6A^2 + B^2)^5 = (6A)^6 \quad \Rightarrow \quad m = 6$

49) $y = 2(3-x)\left(x + \sqrt{x^2+4}\right)$

$\sqrt{x^2+4} + x = t$.....(1),

$$\frac{4}{\sqrt{x^2+4} - x} = t$$

$\sqrt{x^2+4} - x = \frac{4}{t}$.....(2),

Eq (1) – Eq. (2) $\Rightarrow 2x = t - \frac{4}{t}$,

$$y = \left(6 - \left(t - \frac{4}{t}\right)\right)t$$

$= 6t - t^2 + 4 = -9 + 6t - t^2 + 13$
$= 13 - (3 - t)^2 \le 13$

50) Let $P(x) = ax^3 + bx^2 + cx + d$

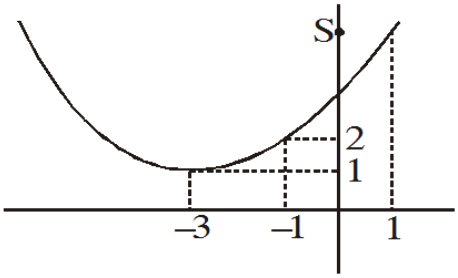
$P(-1) = 10 \Rightarrow -a + b - c + d = 10$ ..... (1)
$P(1) = -6 \Rightarrow a + b + c + d = -6$ ..... (2)
$P(x)$ has max. at $x = -1 \Rightarrow P'(-1) = 0$
$\Rightarrow 3a - 2b + c = 0$ ..... (3)
$P'(x)$ has min. at $x = 1 \Rightarrow P''(1) = 0$
$\Rightarrow 6a + 2b = 0$
Solving (1), (2), (3) and (4), we get
From (4), $b = -3a$
from (3), $3a + 6a + c = 0 \Rightarrow c = -9a$
from (2), $a - 3a - 9a + d = -6 \Rightarrow d = 11a - 6$
from $(1) - a - 3a + 9a + 11a - 6 = 10$
$\Rightarrow 16a = 16 \Rightarrow a = 1$
$\Rightarrow b = -3, c = -9, d = 5$
$\Rightarrow P(x) = x^3 - 3x^2 - 9x + 5$
$P'(x) = 0 \Rightarrow 3x^2 - 6x - 9 = 0$
$\Rightarrow 3(x + 1)(x - 3) = 0$
$\Rightarrow x = -1$ is a point of maxima (given) and $x = 3$ is a point of minima
[$\because$ maxima and minima occur alternatively]

51) $f : [-1, 1] \to R$
$f(x) = ax^2 + bx + c$
$f(-1) = a - b + c = 2$ ...(1)
$f'(-1) = -2a + b = 1$ ...(2)
$f''(x) = 2a$

$\Rightarrow$ Max. value of $f''(x) = 2a = \frac{1}{2}$

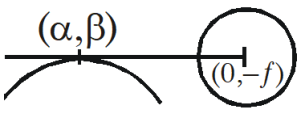
$\Rightarrow$ $a = \frac{1}{4}$; $b = \frac{3}{2}$; $c = \frac{13}{4}$

$\therefore$ $f(x) = \frac{x^2}{4} + \frac{3}{2}x + \frac{13}{4}$



For, $x \in [-1, 1] \Rightarrow 2 \le f(x) \le 5$
$\therefore$ Least value of a is 5

52) $xy' + y = 2x$



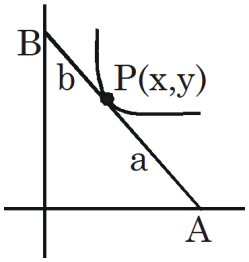
$\Rightarrow \left.\frac{dy}{dx}\right|_{(\alpha,\beta)} = \frac{\beta + f}{\alpha - 0}$

$\Rightarrow \frac{2\alpha - \beta}{\alpha} = \frac{\beta + f}{\alpha} \Rightarrow 2(\alpha - \beta) = f$

⇒ $f.\alpha = 2\alpha^2 - 2\alpha\beta$ (∵ $\alpha\beta = \alpha^2 + 1$)

$= -2$

53) $x^{2017}y^{1033} = K$



$$\frac{dy}{dx} = \frac{-2017y}{1033x}$$

equation of tangent

$$Y - y = \frac{-2017}{1033}\frac{y}{x}(X - x)$$

$$A\left(\frac{3050x}{2017}, 0\right); B\left(0, \frac{3050y}{1033}\right)$$

Let $\frac{a}{b} = \lambda$ ⇒ ratio $\lambda : 1$

$$P\left(\frac{3050x}{2017(\lambda+1)}, \frac{3050\lambda y}{1033(\lambda+1)}\right)$$

$$x = \frac{3050x}{2017(\lambda+1)} \Rightarrow \lambda = \frac{1033}{2017}$$

⇒ $\frac{a}{b} = \frac{1033}{2017}$ ⇒ $a + b = 3050$

sum of digits = 8

54) $x = 3\cos\theta - \cos^3\theta \Rightarrow \frac{dx}{d\theta} = -3\sin^3\theta$

$y = 3\sin\theta - \sin^3\theta \Rightarrow \frac{dy}{d\theta} = 3\cos^3\theta$

$$\frac{dy}{dx} = -\frac{\cos^3\theta}{\sin^3\theta}$$

equation of normal is

$$y - \left(3\sin\alpha - \sin^3\alpha\right) = \frac{\sin^3\alpha}{\cos^3\alpha}\left(x - \left(3\cos\alpha - \cos^3\alpha\right)\right)$$

⇒ $y\cos^3\alpha - x\sin^3\alpha = \frac{3}{4}\sin4\alpha$

⇒ $$4\left[\frac{\cos^4\alpha}{\cos\alpha}.y - \frac{\sin^4\alpha}{\sin\alpha}.x\right] = 3\sin4\alpha$$

 $a = \cos^4\alpha$, $b = \sin^4\alpha$, $a + b = \sin4\alpha$

⇒ $\cos^4\alpha + \sin^4\alpha = \sin4\alpha$

⇒ $2 - \sin^2 2\alpha = 2\sin4\alpha$

⇒ $4 - [1 - \cos4\alpha] = 4\sin4\alpha$

⇒ $3 = 4\cos^4\alpha + 4\sin4\alpha$

⇒ Total 4 values of α